

Mathematics and Numerical Analysis of Deep Learning, Applied to Quantum Many-Body PINNs

Special-curriculum oral exam

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Outline of the talk

1. Terminology: MBSE methods (FCI, HF, CCSD, DMC, VMC, NQS); our ansatz; CTNN.
2. **Paper A** — four questions on deep learning theory.
3. **Paper B** — four questions for PINNs; we focus on Q4 (training).
4. **Our results** — SR+VMC and PINN/collocation; inputs; intrinsic dimension; three-body.

Running thread

The MBSE is nominally infinite-dimensional, but the physical ground state lives on a **low-dimensional manifold**. **Paper A**: when this manifold *exists*. **Paper B**: when it is *reachable*. Our PINN is the worked example.

Terminology and scope

The MBSE problem and the standard solvers

For N electrons in 2D ($d=2N$ configuration space):

$$\hat{H}\Psi = E\Psi, \quad \hat{H} = \sum_i \left(-\frac{1}{2}\nabla_i^2 + \frac{1}{2}\omega^2 r_i^2 \right) + \sum_{i<j} \frac{1}{r_{ij}}.$$

- HF** mean-field; one Slater determinant; misses correlation.
- CCSD** truncated cluster expansion; scales polynomially; fails in strong-correlation regimes (Wigner).
- FCI** exact in a basis; basis dimension grows combinatorially in d .
- DMC** stochastic projection $\propto e^{-\tau\hat{H}}$ onto the ground state; fixed-node bias.
- VMC** sample $|\Psi_\theta|^2$, evaluate $\langle\hat{H}\rangle$; minimize variationally.
- NQS** neural ansatz used inside VMC (Carleo–Troyer 2017 onward).

What ML buys: expressivity *and* scaling — FCI has the first, CCSD the second. The bet is that a neural ansatz can have both.

Our ansatz

$$\Psi_{\theta,\beta}(R) = \text{SD}(\tilde{R} + \Delta_{\beta}(\tilde{R})) \exp\left(\underbrace{\sum_{i<j} u_{\sigma_i\sigma_j}(\tilde{r}_{ij})}_{\text{analytic cusp}} + \underbrace{W_{\theta}(\tilde{R})}_{\text{neural correlator}}\right).$$

Pieces.

- **Slater determinant**: antisymmetry.
- **Analytic cusp**: $1/r$ Kato condition; removes the singular part from E_L .
- **Neural correlator** W_{θ} : three Deep-Sets-style branches (ϕ, ψ, ρ) .
- **Backflow** Δ_{β} : a CTNN; near-identity ODE flow.

Trap units $\tilde{r} = \sqrt{\omega} r$: makes the harmonic length $\ell \rightarrow 1$ across ω .

Why the determinant sees backflow but the correlator does not. Keeps the cusp/Jastrow on the unmodified coordinates — prevents a self-reinforcing feedback into the local energy.

Why three networks. Particle-only Deep Sets cannot represent pair correlation (Haas et al.); a pair branch is mandatory.

CTNN architecture

One round of message passing on the particle graph:

$$\mathbf{m}_{ij} = \phi_{\beta}(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j, \mathbf{r}_{ij}, \dots) \cdot w_{ij}^{\text{spin}}, \quad \mathbf{m}_i = \sum_{j \neq i} \mathbf{m}_{ij}, \quad \delta \tilde{\mathbf{x}}_i = \psi_{\beta}(\tilde{\mathbf{x}}_i \| \mathbf{m}_i),$$

$$\Delta_{\beta} = \tanh(\delta \tilde{X}) \cdot s_{\beta}, \quad s_{\beta} = \text{softplus}(s_{\beta}^{\text{raw}}) > 0, \quad \text{last layer of } \psi_{\beta} \text{ zero-initialized.}$$

Why these choices.

- Permutation-invariant (sum aggregation).
- Rotation-equivariant (vector messages).
- Near-identity at init $\Rightarrow$ ResNet/ODE-flow regime (Paper A §1.6.2).
- $k=1$ round of MP $\Rightarrow$ up to *three-body* correlations.

Empirical guarantee. $\|\Delta_{\mathbf{x}}\|/\ell \lesssim 0.10$ in production; conventional BackflowNets can reach ~ 0.95 in poorly-converged regimes.

Paper A — Modern Mathematics of Deep Learning

Paper A — the four organizing questions

Berner, Grohs, Kutyniok, Petersen (2021).

$$R(f_S) - R(f^*) = \underbrace{R(f_S) - R(f_{\mathcal{F}})}_{\text{generalization}} + \underbrace{R(f_{\mathcal{F}}) - R(f^*)}_{\text{approximation}}.$$

Classical bounds (VC / Rademacher) need $\dim(\mathcal{F}) \ll n$. Deep nets have parameters $\gg$ data, fit the training set exactly, yet generalize. *Classical bounds are vacuous.*

- **Q1** How do over-parameterized NNs *generalize*?
- **Q2** What role does *depth* / width play in expressivity?
- **Q3** How do NNs beat the *curse of dimensionality*?
- **Q4** Why does *non-convex optimization* succeed?

A–Q1: Generalization

The phenomenology is robust; no unified theory.

- **Double descent** (Belkin et al. 2019): past the interpolation threshold, test error falls a *second* time. Empirical fact the classical U-curve cannot accommodate.
- **NTK regime** (Jacot 2018, very wide nets): training linearizes into kernel regression with the Neural Tangent Kernel $\Theta(x, x') = \langle \nabla_{\theta} f(x), \nabla_{\theta} f(x') \rangle$ frozen at init; convergence set by its spectrum.
- **For us:** the SR matrix $S = \text{Cov}_{|\Psi|^2}[\partial_{\theta} \log |\Psi|]$ is exactly this NTK (of $\log |\Psi|$) in parameter space.

Honest verdict (article, §1.2): robust phenomenology, no unified theory.

A-Q2: Expressivity — depth and width

Universal approximation (Cybenko 1989, Hornik 1991): one hidden layer with non-poly. activation is dense in $C(K)$ — a *topological* statement, no rate.

Width vs depth (article example). Approximating $\|x - y\|$ on $[0, 1]^d \times [0, 1]^d$ to error ε :

1 hidden layer: width $\Omega(2^{cd})$, multi-layer: width $\text{poly}(d)$.

Depth provides *categorically* larger function classes at fixed budget.

What width is for (article). With ReLU, width $\geq d_{\text{in}} + 1$ is necessary for universality at all (Hanin–Sellke); width controls the dimension of the manifold along which the function can vary smoothly. Depth controls compositionality; width controls cross-section.

Depth is how NNs realize hierarchical / compositional structure that classical (sparse-grid, spectral) methods cannot.

A–Q3: Curse of dimensionality — Barron, intrinsic d

Curse: approx. C^s function on $\mathbb{R}^d$ to error ε needs $\sim \varepsilon^{-d/s}$ neurons — exponential in d . (s = smoothness order; the more derivatives the target has, the faster classical methods converge.)

Barron's theorem. For g with $C_g = \int_{\mathbb{R}^d} \|\xi\|_2 |\hat{g}(\xi)| d\xi < \infty$, a two-layer net with n neurons satisfies

$$\inf_{\theta} \|\Phi_{(d,n,1),\varrho}(\cdot, \theta) - g\|_{L^2(\mu)} \leq \frac{c C_g}{\sqrt{n}}.$$

Exponent **independent of d** . Dimension hidden in C_g . Proof: Monte Carlo over n random neurons.

Connection to MBSE. **HF / FCI** take a $d=40$ problem and expand against a basis that grows combinatorially. **CCSD** keeps polynomial scaling by sacrificing expressivity — collapses in the Wigner regime. **NN ansatz**: find the *intrinsic* dimension.

Note (§1.4.3): Coulomb / cusp parts of Ψ are *not* in Barron space; we *condition* the problem (cusp factor, soft features) to push the learned part into a Barron-friendly class.

A–Q4: Optimization — gradients, NTK, SR

The puzzle. The loss is non-convex with exponentially many critical points; worst-case theory says SGD should fail. It doesn't. Why?

- **Landscape & saddle escape:** critical points are dominated by saddles, not bad local minima (Dauphin 2014); the stochastic noise in SGD kicks the iterate off saddles.
- **Implicit regularization:** among the many interpolating solutions, (S)GD from small init drifts to the min-norm / flat one $\Rightarrow$ good generalization for free.
- **Adam:** per-coordinate adaptive step; diagonal preconditioner from the running second moment of the gradient. Robust, axis-aligned curvature only.
- **Natural gradient:** $F^{-1}\nabla L$ with Fisher F . **Quantum Fisher = NTK of $\log |\Psi|$ = Stochastic Reconfiguration S matrix** — our production optimizer.

Architecture as optimization aid. ResNets = Euler step of an ODE $\Rightarrow$ cures vanishing/exploding gradients (our backflow is a near-identity flow); CNN/GNN equivariance shrinks the function class.

Paper B — Numerical Analysis of PINNs

Paper B — spine, Q1–Q3 in one slide

De Ryck & Mishra (2024).

$$\|u_\theta - u\|_{\mathcal{X}} \leq C_{\text{stab}} \left(\underbrace{\mathcal{E}_{\text{approx}}}_{\text{Q1}} + \underbrace{\mathcal{E}_{\text{gen}}}_{\text{Q3}} + \underbrace{\mathcal{E}_{\text{opt}}}_{\text{Q4}} \right).$$

- **Q1 — approximation.** tanh-nets reach Sobolev rate $\varepsilon^{-d/(k-j)}$; this is just universal approximation in H^k . For us **Paper A** suffices.
- **Q2 — stability.** Assume a low generalization loss; *then* low training loss bounds the solution. **For us:** zero-variance principle. $\text{Var}[E_L] = 0 \iff \Psi$ is an exact eigenstate; both losses vanish together.
- **Q3 — generalization (quadrature).** Concentration: $\sup_\theta |L_{\text{train}} - L_{\text{true}}| \lesssim \text{Lip} \sqrt{\log \mathcal{N}/n}$. **For us:** always evaluated by an independent heavy VMC.

Paper B's headline: the binding bottleneck is *training*, and it is governed by the spectrum of the *Hermitian square* L^*L .

B-Q4: Training = conditioning of $L^*L \circ TT^*$

Linearize GD around θ_0 : $q_i = \partial_{\theta_i} u_{\theta_0}$.

$$A_{ij} = \langle Lq_i, Lq_j \rangle, \quad \#(\varepsilon) = O(\kappa(A) \ln \frac{1}{\varepsilon}), \quad \kappa(A) = \lambda_{\max}/\lambda_{\min}.$$

Theorem 7.10. $\kappa(A) = \kappa(\mathbf{A} \circ TT^*)$, with $\mathbf{A} = L^*L$ (operator stiffness from the PDE) and TT^* the NTK as an integral operator (network curvature). Product = stiffness GD feels.

Laplacian. $L = -\Delta \Rightarrow L^*L = \Delta^2$; Fourier symbol $|k|^4 \Rightarrow \kappa \sim (k_{\max}/k_{\min})^4$. Supervised learning is the special case $L=\text{Id}$, $\mathbf{A}=I$. All extra PINN difficulty is in L^*L .

Preconditioning: choose P with $PP^\top \approx (L^*L)^{-1}$.

Catch-22: TT^* -preconditioning is not enough when L^*L is stiff

Setting. $N=6, \omega=0.5$, backflow ansatz, collocation training. **17 interventions** (hard mining, smoothness penalty, SIR, Gumbel-top-K, hybrid loss, det filter, soft Coulomb, kinetic clipping, K-FAC, Adam variants, ...): all give error in $[0.32, 24.7]\%$. **Same ansatz, VMC+SR:** $< 0.01\%$.

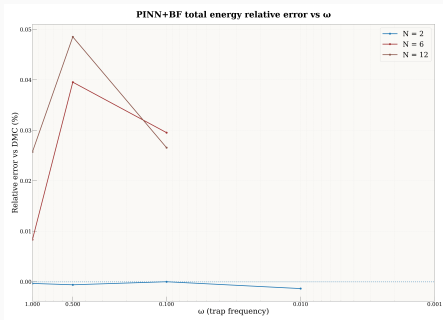
What helped. Removing the Laplacian path from backflow gradients (REINFORCE only) while keeping the kinetic term in the loss. L^*L no longer couples to backflow parameters.

Theorem 7.10 in negative form. K-FAC / Adam precondition TT^* (parameter space); stiffness sits in L^*L . *Necessary but not sufficient.* Changing the form (variational, REINFORCE) removes the stiff piece.

Why the analytic cusp works. Kato slope fixed analytically $\Rightarrow$ the network never represents $1/r$ in E_L ; a non-Barron feature taken out of the function class entirely.

Our results

SR+VMC results: PINN+CTNN matches DMC across the phase diagram



Relative error vs. ω (SR+VMC).

N	ω	DMC	PINN+BF	PINN+CTNN
2	1.0	3.00000	-0.0003%	-0.0000%
6	0.1	3.55385	+0.0295%	+0.0008%
6	0.5	11.78484	+0.0395%	-0.0000%
6	1.0	20.15932	+0.0083%	-0.0041%
12	0.5	39.1596	+0.0485%	+0.0018%
12	1.0	65.7001	+0.0257%	-0.0069%
20	0.5	93.8752	—	+0.0040%
20	1.0	155.8822	—	-0.0024%

CTNN cell ablation, $N=6$: $\omega=1$: + 22.4%, $\omega=0.1$: + 30.5%, $\omega=0.001$: + 11.8%. *Entire increase is kinetic* $\Rightarrow$ CTNN fixes nodal geometry, not amplitudes.

PINN/collocation results: MCMC-free training of the CTNN

N	ω	PINN-only (best)	vs DMC
6	1.0	20.1613(23)	+0.013%
6	0.5	11.7847(20)	+0.002%
6	0.1	3.5571(4)	+0.263%
6	0.01	0.69141(8)	+0.237%
6	0.001	0.14118(1)	+0.250%
12	1.0	65.712(4)	+0.018%
12	0.5	39.171(4)	+0.028%
20	1.0	158.14(2)	+1.45% ^J
20	0.5	96.45(3)	+2.74% ^J

^JJastrow-only beats BF at $N=20$ (memory scaling).

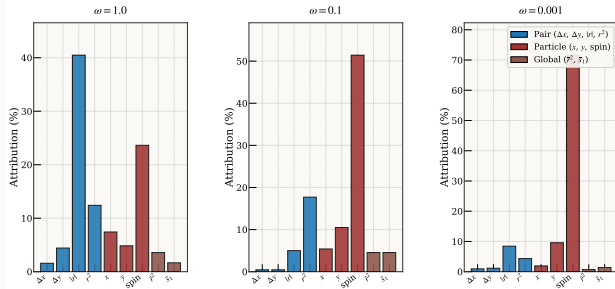
Tricks that mattered.

- Three-stage cascade (REINFORCE $\rightarrow$ low-LR $\rightarrow$ hard-focus).
- Importance-resample 4096 from $8\times$ oversampled mixture proposal.
- REINFORCE only (no ∇_{θ} through the Laplacian).
- Rollback gate on per-epoch variance.

Where SR breaks down. $\omega \leq 0.01$: Fisher matrix becomes rank-deficient (eigenvalues span many decades); SR gives +4–36%, Adam recovers +0.15–0.24%.

Inputs the CTNN uses — attribution across regimes

Input channel attribution of the Jastrow correlator



$N = 6$. $\omega \in \{1.0, 0.1, 0.001\}$.

Channels. 6 safe radial; raw ($\Delta x, \Delta y$); one-body (x, y); spin; 2 global stats.

Observed.

- $\omega=1.0$: $|r|$ dominates (40.5%).
- $\omega=0.1$: spin 51.4%.
- $\omega=0.001$: spin 71.6% (Wigner).
- Raw ($\Delta x, \Delta y$) suppressed in every regime.

The network self-selects bounded, informative channels in each regime — a **conditioning device**. This is the **Paper B** principle (slide 12) realized on the input side: well-conditioned features lower κ along the trajectory. The shift radial→spin tracks the physical Wigner crossover.

Intrinsic dimension — the manifold the network finds

Effective rank $r_{\text{eff}}(Z) = \exp(-\sum_i p_i \log p_i)$, $p_i = \sigma_i^2 / \sum \sigma_j^2$, on the correlator features $Z = \bar{h}^\phi \| \bar{h}^\psi \| \mathbf{g}$.

N	ω	r_{eff}	$ \rho $	PC1 leader
2	0.001	1.00	1.00	ψ
2	1.0	1.53	0.98	$\mathbf{g}$
6	0.001	1.52	0.97	$\mathbf{g}$
6	0.1	2.73	0.94	ϕ
6	1.0	2.69	0.99	ψ
12	0.001	1.66	0.96	ϕ
12	0.1	3.08	0.99	ψ
12	1.0	1.76	0.99	ψ
20	0.001	2.59	0.99	ψ
20	1.0	2.32	0.99	ψ

What it says.

- $r_{\text{eff}} \leq 3.1$ on the whole (N, ω) grid — even for $d=40$ ($N=20$).
- Head reconstructed from PC1 alone: $|\rho| > 0.94$ for $N \geq 6$.
- The leading axis *rotates* across regimes ($\phi \leftrightarrow \psi \leftrightarrow \mathbf{g}$); the manifold does not *grow*.

For backflow. Different object: $\text{rank} \approx 2N - 2$ (full geometric), CKA with correlator > 0.98 everywhere $\Rightarrow$ acts in the orthogonal complement.

Scaling. r_{eff} stays $O(1)$ as N grows from 2 to 20. The physics is low-dimensional regardless of ambient d .

Three-body sensitivity — what one extra round of MP buys

Definition. Bin by pair distance. Pairwise-only f has zero intra-bin variance by construction; the CTNN cell can have variance from beyond-pairwise structure (a third particle changes the message to i even when r_{ij} is fixed). Ratio:

$$r(\omega) = \text{Var}_{\text{intra}}[\text{CTNN cell}] / \text{Var}_{\text{intra}}[\text{pairwise}].$$

Data ($N=6$). $r(1.0) = 2.03$, $r(0.1) = 2.93$, $r(0.001) = 1.05$.

Interpretation. At moderate-to-weak correlation the cell genuinely needs three-body content; in deep Wigner the structure is set by the classical lattice (pairwise), capacity present but unused.

Codimension-in-pairwise: $r(\omega) \rightarrow 1$ when the optimal correlator lies in $\mathcal{F}_0$; $r(\omega) > 1$ otherwise. A variance ratio measuring the codimension of the physical solution inside the architectural function class.

- **Paper A:** the manifold *exists* — Barron, depth, compositionality, symmetry break the curse; architecture encodes the symmetry.
- **Paper B:** the manifold is *reachable* iff $\kappa(L^*L \circ TT^*)$ is controlled — nearly every successful trick is preconditioning.
- **Our PINN** sits at the intersection: conditioning (cusp, soft features, trap units, REINFORCE, SR/Adam regime split) brings $r_{\text{eff}} \leq 3$ within reach for $d = 40$.

Take-away. The many-body wavefunction lives on a 1–3-dimensional manifold that a symmetry-equivariant, well-conditioned network can reach with $\sim 3 \times 10^3$ MCMC samples per SR step — orders of magnitude better than basis blow-up implies.

Backup: NTK proof sketch and the linearization

$f_t(x) = f_{\theta_t}(x)$. Chain rule on $\dot{\theta} = -\nabla_{\theta} L$:

$$\dot{f}_t(x) = -\sum_{x'} \Theta_t(x, x') (f_t(x') - y(x')), \quad \Theta_t(x, x') = \sum_p \partial_{\theta_p} f_t(x) \partial_{\theta_p} f_t(x').$$

Jacot. As width $m \rightarrow \infty$ with appropriate scaling: parameters move only $O(1/\sqrt{m})$ while the function moves $O(1)$, so $\Theta_t \rightarrow \Theta_0 = \text{const.}$ Training linearizes; dynamics governed by $\text{spec}(\Theta_0)$.

Identification with SR. $S_{ij} = \text{Cov}_{|\Psi|^2}[O_i, O_j]$, $O_i = \partial_{\theta_i} \log |\Psi|$. S is the quantum Fisher matrix = NTK of $\log |\Psi|$ in the sampling measure. SR update $S \delta\theta = -g$ is natural gradient = explicit TT^* preconditioner.

Backup: Barron proof sketch

$g \in L^1(\mathbb{R}^d)$ with $C_g = \int \|\xi\|_2 |\hat{g}(\xi)| d\xi < \infty$. Write

$$g(x) - g(0) = \int_{\mathbb{R}^d} (e^{i\xi^\top x} - 1) \hat{g}(\xi) d\xi.$$

Re-express as $C_g \mathbb{E}_{(\Xi, \tilde{\Xi}) \sim \mu_g} [\Gamma(\Xi, \tilde{\Xi})(x)]$ with Γ of fixed form. Average n iid draws; Bienaymé's identity gives variance $\leq (2\pi C_g)^2/n$, so the average is within $2\pi C_g/\sqrt{n}$ in $L^2(\mu)$. Each Γ is well-approximated by a single neuron. $\Rightarrow$ two-layer net of size n at MC rate.

The neurons are Monte Carlo samples in frequency space.

Backup: design choices $\rightarrow$ preconditioners

Technique	What it conditions	Effect on κ
Trap units $\tilde{r} = \sqrt{\omega}r$	rescales $\ell \rightarrow O(1)$ across ω	shrinks $k_{\max}/k_{\min}$
Soft-core feats $ds_k/d\tilde{r} \rightarrow 0$	caps high- $ k $ in learned part	bounds $k_{\max}$
Analytic cusp $\gamma r e^{-r}$	cancels $1/r$ residual spike	removes worst $ k $
Variational / REINFORCE	lowers operator order	replaces L^*L stiffness
Stochastic Reconfiguration	explicit $P \approx \mathbf{A}^{-1/2}$ (Fisher)	$\kappa \rightarrow O(1)$ in good regime

Local energy + REINFORCE.

$$E_L = -\frac{1}{2}(\Delta \log |\Psi| + \|\nabla \log |\Psi|\|^2) + V, \quad \nabla_\theta E = 2 \mathbb{E}[(E_L - b) \nabla_\theta \log |\Psi|].$$

Differentiates only through the score $\Rightarrow$ no ∇_θ through the Laplacian path $\Rightarrow$ resolves the backflow catch-22.

Backup: 2D cusp + log-divergent squared spike

Antiparallel (no node): $\partial_r \log \Psi|_0 = 1$. Parallel (linear node, 2D, $k=1$): $\partial_r \log J|_0 = \frac{1}{2k+d-1} = \frac{1}{3}$.

A cusp-slope error δa gives $E_L \supset -\delta a/r$. Squaring:

$$\int_{r < \varepsilon} (1/r)^2 d^2 \mathbf{r} = +\infty \quad (\text{log-divergent in 2D}).$$

Strong residual dominated by close-pair samples $\Rightarrow$ heavy-tailed MC. Our analytic cusp plus soft-core features make E_L bounded $\Rightarrow$ Paper B's quadrature bounds (§6) apply.

Backup: backflow as a force-aligned object across the Wigner crossover

For each sample, take median $\cos(\Delta \mathbf{x}, \mathbf{F})$ against trap, Coulomb, total classical force. Result for $N=6$:

ω	$\cos(\Delta, \mathbf{F}^{\text{trap}})$	$\cos(\Delta, \mathbf{F}^{\text{Coul}})$	$\cos(\Delta, \mathbf{F}^{\text{tot}})$
1.0, 0.1	+0.97, +0.98	-0.90, -0.93	$\approx +0.97$
0.001 (Wigner)	-0.83	+0.83	≈ -0.6

Sign flip across the Wigner crossover. In the fluid regime backflow opposes Coulomb (over-repulsion correction). At Wigner it switches: it now follows Coulomb and opposes the trap, i.e. becomes a lattice-correction mode against the trap's isotropic restoring force. The diagnostic matches the Lindemann / topology-collapse signatures of the Wigner crossover.

Backup: sharp PINN–CTNN coupling transition at $\omega \approx 0.1$

$\langle \cos(\nabla_x f, \Delta x) \rangle$ and BF-induced output variance $|\delta f|/\sigma_f$:

N	ω	$\langle \cos \rangle$	$ \delta f /\sigma_f$
6	0.01	0.18	0.05
6	0.1	0.49	0.25
6	1.0	0.77	0.41
12	0.001	-0.48	0.03
12	0.1	0.56	0.39
20	1.0	0.78	0.69

Below $\omega \approx 0.1$: CTNN works in the orthogonal complement of $\nabla_x f$, modifying only the determinant; barely affects f . Above 0.1: cooperative refinement; alignment $\rightarrow 1.0$; BF accounts for up to 70% of the correlator output. The transition coincides with the physical crossover in $\Gamma = V_{\text{int}}/T$.

Backup: $N=20$ — Jastrow-only beats backflow

Best collocation results at $N=20$ ($d=40$):

$$\omega=1.0 : +1.45\% \text{ (J-only)} \quad \text{vs} \quad +18\% \text{ (BF)}.$$

Why. BF edge features scale $\mathcal{O}(N^2 h_{\text{bf}})$; at $N=20$ this forces a collocation-budget / hidden-dim cut that destroys gradient quality more than the extra expressivity compensates. *Architectural expressivity must be paid for in gradient-estimation budget*; the SR / ESS analysis predicts exactly where the trade reverses.